Automated Gearbox

Jan-Feb 2026 Personal Project

Overview

1st gear (Driver)

Module: 1.0

PCD: 20 mm

Teeth: 20

Face width: 10 mm

(8-12 x m)

Root fillet radius: 0.2-0.4 mm
0.3

2nd gear (driven) w/ 5:1 reduction

Module: 1.0

PCD: 100 mm

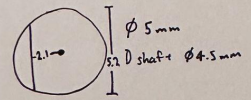
Teeth: 100

Face width: 10 mm

RFR: 0.2-0.4 mm
0.3

Ø Hole: ~~6.35 + 2(0.05) mm~~
+ clearance
7.9375 + 2(0.10) mm

Backlash: 0.2 mm



Housing

Center distance for mounting $= \frac{PCD_1 + PCD_2}{2}$

$= \frac{20 + 100}{2} = 60$

Minimum internal ~~height~~ width: clearance + ~~Ø6.35~~ thickness + clearance
~~5 + 2(1.35) mm + 5 = 15.35 mm~~
5 + 10 + 5 = 20 mm

Minimum internal length: clearance + ~~thickness~~ $\frac{1}{2}$ Ø0.1 + center distance + ~~thickness~~ $\frac{1}{2}$ Ø0.2 + clearance

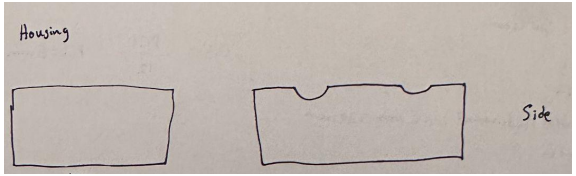
$= 5 + \frac{1}{2}(20) + 60 + \frac{1}{2}(100) + 5 = 130$ mm

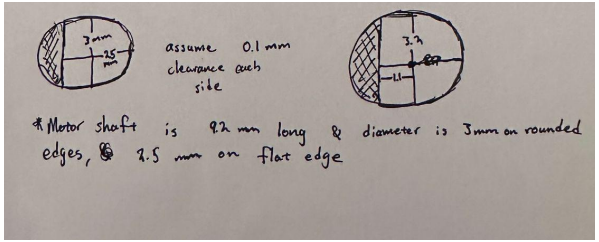
Minimum depth: clearance + $\frac{1}{2}$ Ø0.2
$= 10 + \frac{1}{2}(100) = 60$

Wall thickness: 5 mm

Shaft hole 1: allow space to rotate Ø6 mm

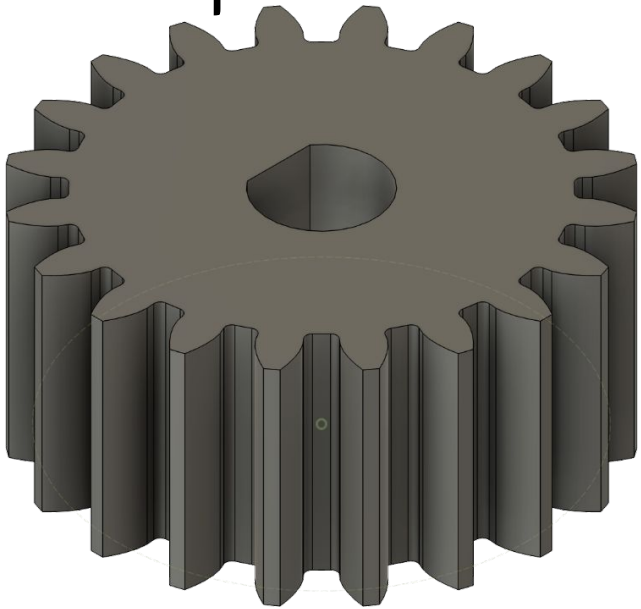
Shaft hole 2: Ø9 mm



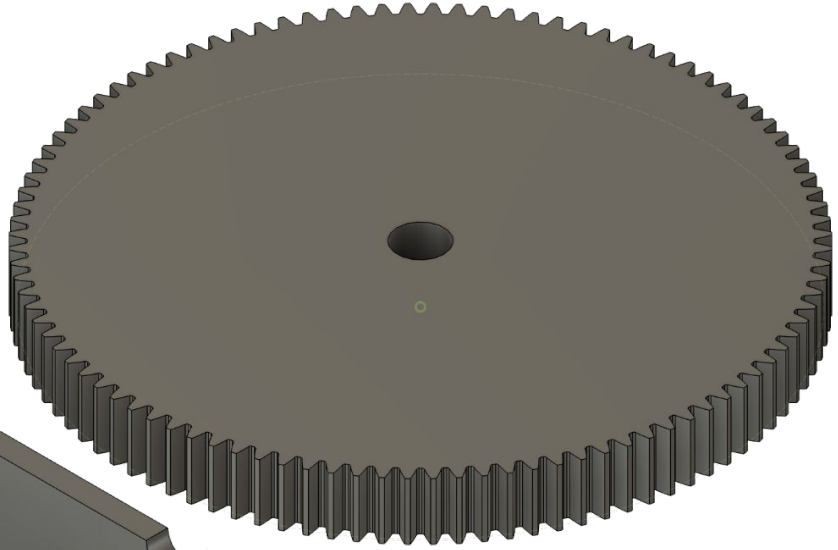


*Motor shaft is 9.2 mm long & diameter is 3mm on rounded edges, 2.5 mm on flat edge

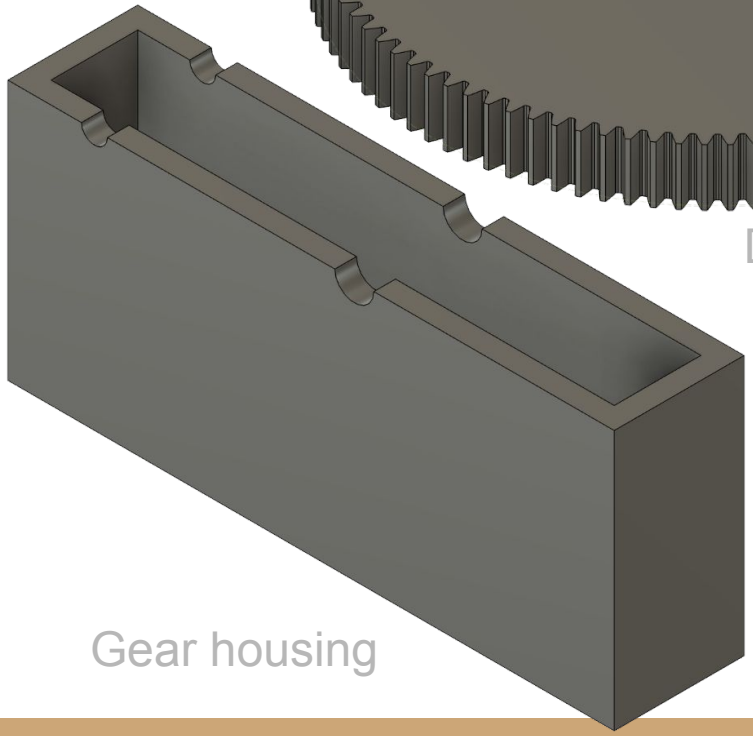
CAD files



Driver gear (20T)



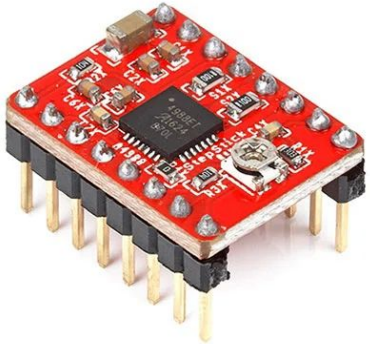
Driven gear (100T)



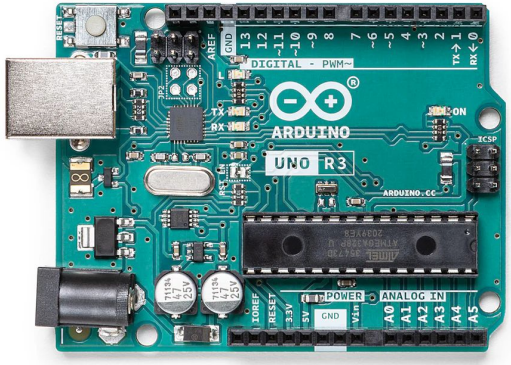
Gear housing

Electronic components

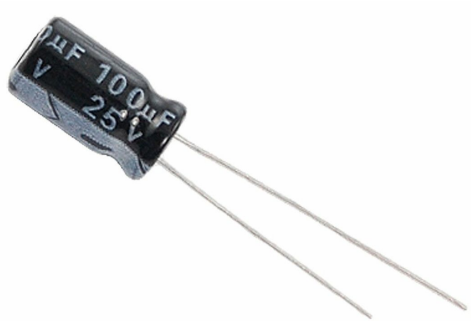
Nema 17 Stepper Motor



A4988 Stepper Motor Driver

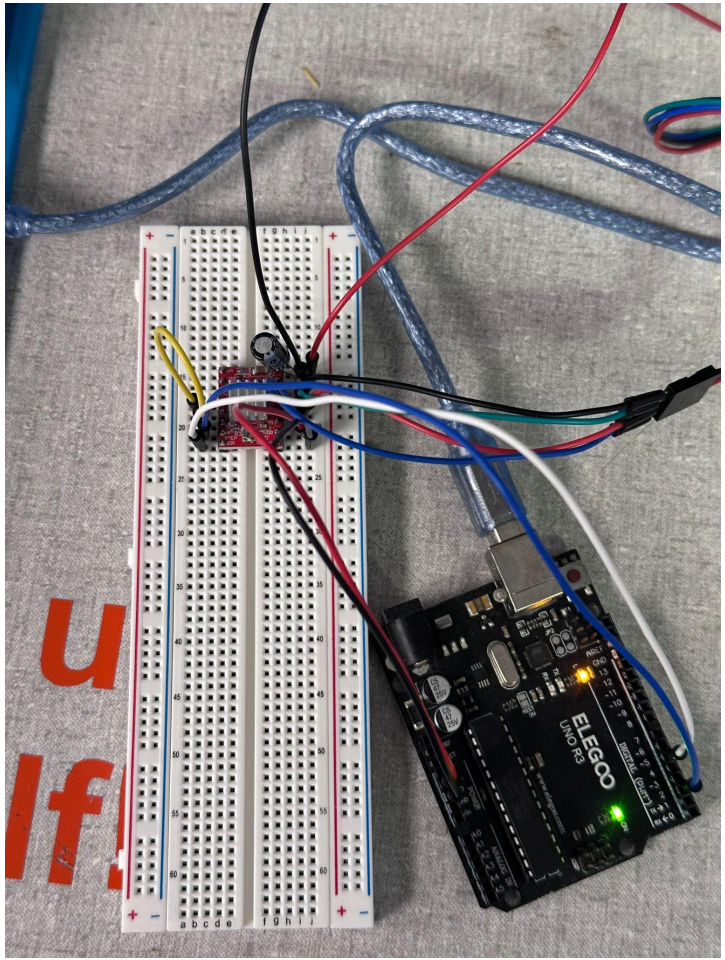


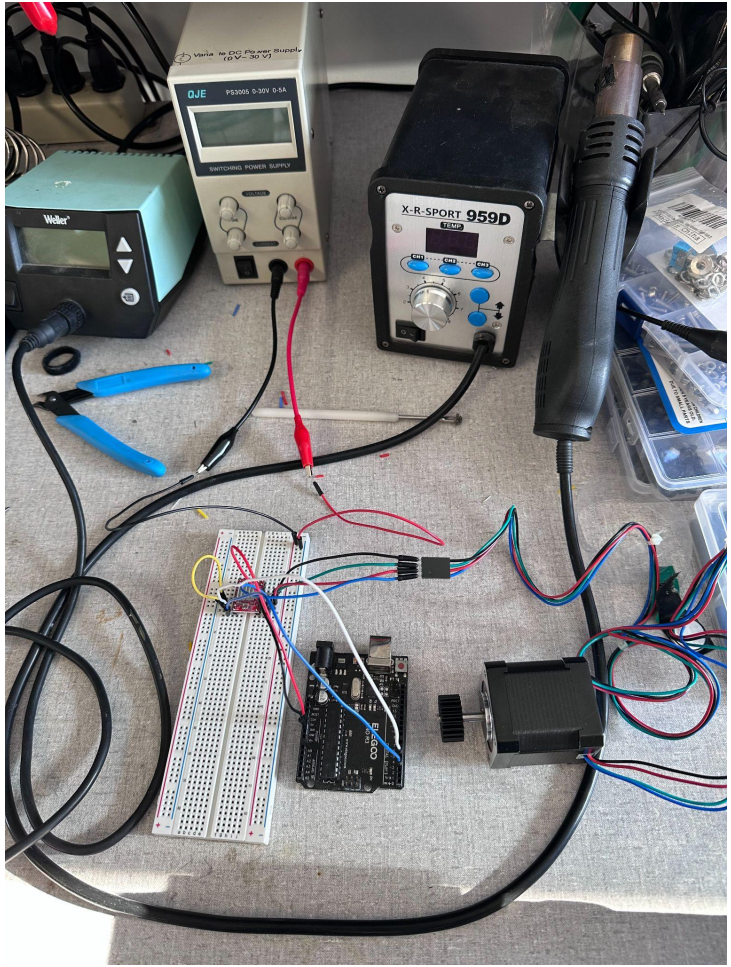
Arduino Microcontroller



100 µF capacitor

Wiring


ELEGOO
UNO R3


QJE
PS3005 0-30V 0-5A
SWITCHING POWER SUPPLY
Weller
X-R-SPORT 959D
TEMP
ELEGOO

Motor Test

Outcome